

EEG Projects

BioRTC 2026 · Yobe State University

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Project 1: Seizure Detection using Deep Learning

The problem

Epilepsy affects 50 million people worldwide. 30% do not respond to medication.

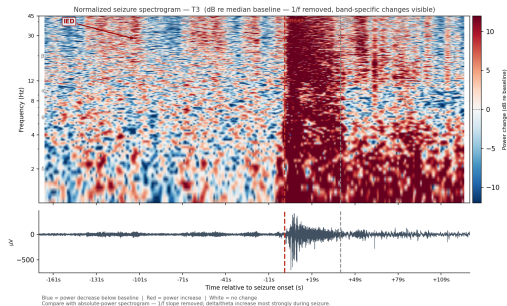
Automated seizure detection is a step towards:

- Accurate seizure diaries (replacing unreliable self-report)
- Real-time alerts for caregivers
- Objective pre-surgical evaluation

Dataset: TUSZ v2.0.6

3 000+ patients, expert seizure labels,

<5% of windows are seizures, 95% accuracy = detect nothing



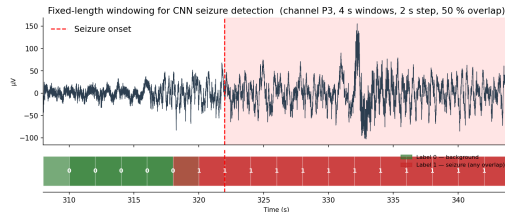
Normalized spectrogram (dB re baseline), channel T3. Red = power increase; blue = decrease.

White vertical lines = seizure onset/offset. Post-ictal IEDs visible as bright vertical streaks.

Project 1: What You Will Do

Step 1 - Offline detection

- Preprocess TUSZ recordings
- Segment into 4 s windows, extract band-power features
- Train an **MLP** on features
- Train a **1D-CNN** on the raw signal
- Compare sensitivity / specificity / F1



4 s windows, 50% overlap. Red = seizure label. Note: seizure windows are a small fraction of the total.

Step 2 - Online detection

- Re-split by **time** (not randomly)
- Sliding window: latency to first correct detection
- Why does random splitting give an unfair advantage?

Track: CS / data science **Group:** 2-4 students

Notebooks: EEG_01 + EEG_02

Step 3 - Transfer to Yobe

Run on SUBJECT1.edf. No labels. Compare against the clinical report.

Project 2: Source Reconstruction

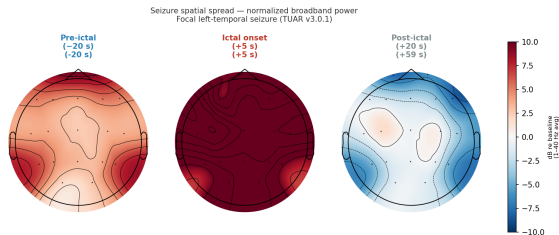
The problem

Each electrode picks up a **mixture** of signals blurred by the skull.

You cannot tell from electrode amplitudes alone **where a seizure originates**.

Why this matters

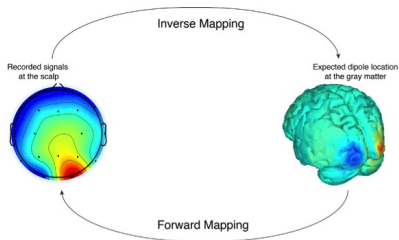
Understanding where a seizure starts, how it spreads - is a fundamental open question in epilepsy research. Different inverse methods make different assumptions and do not always agree, making the comparison itself scientifically interesting.



Pre-ictal: resting baseline. Ictal: broadband power increase.

Post-ictal: power reduction. The scalp maps show *what* changes - but not *where in the brain* it originates.

Project 2: The Inverse Problem



Forward problem (left): one dipole gives one unique scalp map. Inverse problem (right):

the same scalp map is consistent with many possible source configurations.

Three methods, one seizure

- **dSPM** - noise-normalized minimum norm; good spatial coverage
- **sLORETA** - zero localization error for point sources
- **LCMV beamformer** - suppresses activity from other locations

Questions to answer

- Do all three methods point to the **same region**?
- If they disagree - which do you trust, and why?
- How does the seizure **spread** over time?
- Does the focus match **channel T3** (the electrode showing the seizure first)?

Project 2: What you will use

Dataset: TUSZ v2.0.6

Same recordings as Project 1.

Standard **fsaverage** head model (no individual MRI needed).

	Project 1 - DL Seizure Detection	Project 2 - Source Reconstruction
Track	CS / data science	Computational neuroscience
Dataset	TUSZ v2.0.6	TUSZ v2.0.6
Methods	MLP, 1D-CNN	dSPM, sLORETA, LCMV
Challenge	Class imbalance, online evaluation	Ill-posed inverse problem
Extension	Transfer to Yobe EEG	Localize in Yobe EEG
Notebooks	EEG_01 + EEG_02	EEG_01 + EEG_03

Both projects use the **same TUSZ recordings** and the same MNE-Python preprocessing pipeline.

Both end with an **open question**: does your method transfer to a real hospital recording from Yobe with no ground truth?